

Linux environment and the ATmega microcontroller. The Bridge facilitates this communication. The diagram shows how various peripherals are connected via the bridge.

The diagram shows that the Arduino™ environment codes are executed on the ATmega 32u4 microcontroller stored via the USB programmer. The bridge acts as an interface between the ATmega 32u4 microcontroller and the Atheros AR9331 processor. The Atheros AR9331 processor can connect to Wi-Fi and Ethernet network, store data in an SD card and also act as USB host to connect peripheral devices such as camera, phones, keyboard, mouse etc.

The bridge empowers the Arduino™ sketches (programs) with the power to run Linux commands in the Linux environment. Thus, the bridge allows the running of shell scripts and transmission of data to and from the Atheros AR9331 processor. It thereby allows indirect use of communication interfaces such as Wi-Fi and Ethernet for the ATmega 32u4 microcontroller. Similarly, it allows indirect use of the SD card and the USB host interface. Thus, the Arduino™ sketches can use peripheral devices attached to the USB host.

Because of the use of two processors, pins on the Arduino™ Yún board are built to behave in a slightly different manner



Serial communication port

- ◆ **Two Wire Interface (TWI communication):** 2 (SDA) and 3 (SCL) pins are used for TWI communication. The Wire library supports communication via the TWI interface.
- ◆ **External Interrupts:** Pins 3 (interrupt 0), 2 (interrupt 1), 0 (interrupt 2), 1 (interrupt 3) and 7 (interrupt 4) can be configured to trigger an interrupt under various conditions. As pins 0 and 1 talk to the Linux processor, they should be avoided for use as interrupt.
- ◆ **PWM:** 3, 5, 6, 9, 10, 11 and 13. Provide 8-bit PWM output with the 'analogWrite()' function.
- ◆ **SPI:** On the ICSP header. SPI communication is supported via the SPI library. Be careful when using shields that use SPI, as it works with slight difference on the Yún.
- ◆ **Yún RST:** A LOW signal on these pins will reset the AR9331 micropro-